

Gears

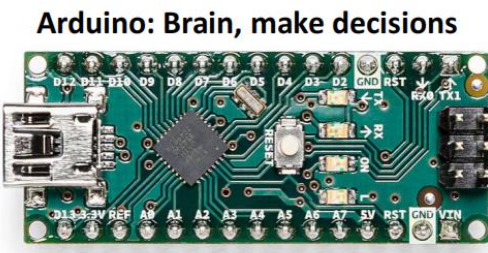
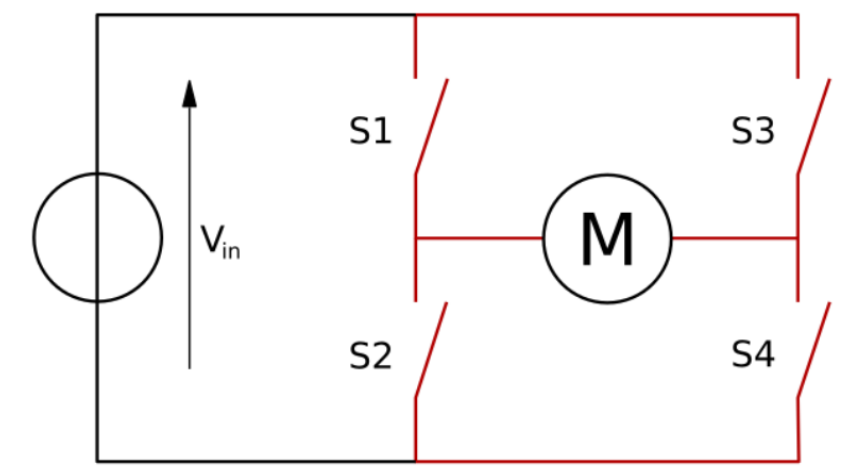
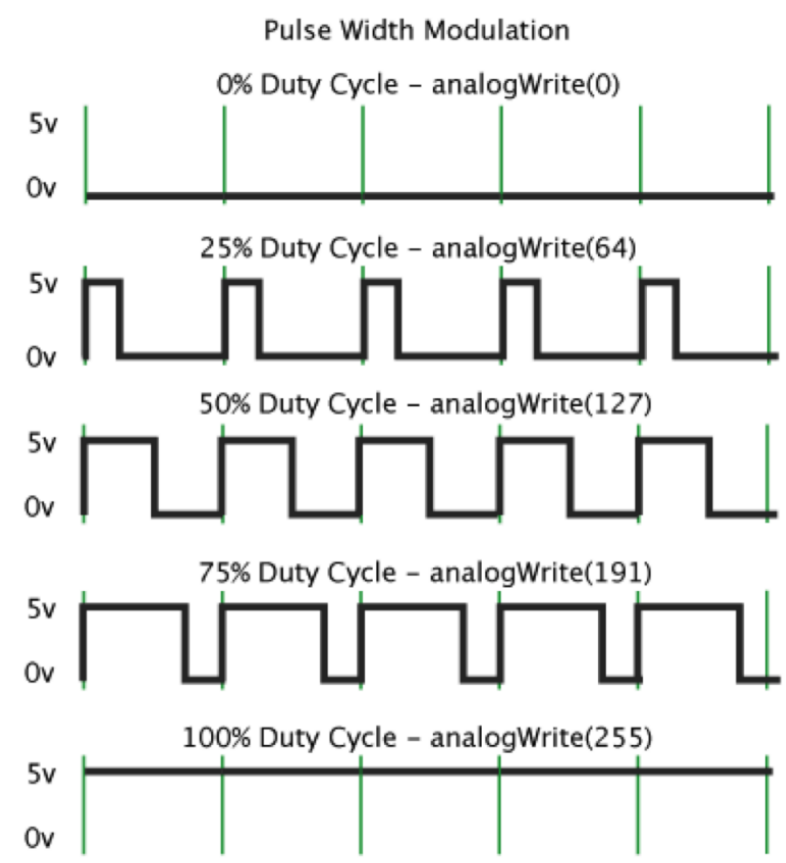
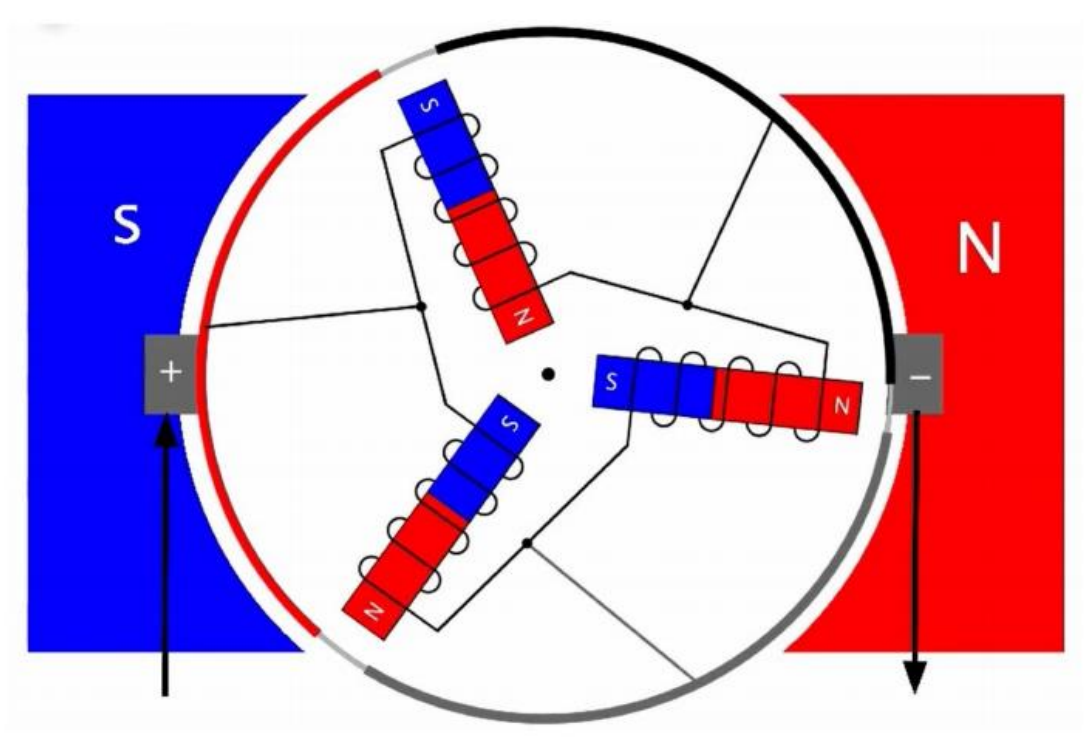
MEGN 441, Lecture 3

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Last Time

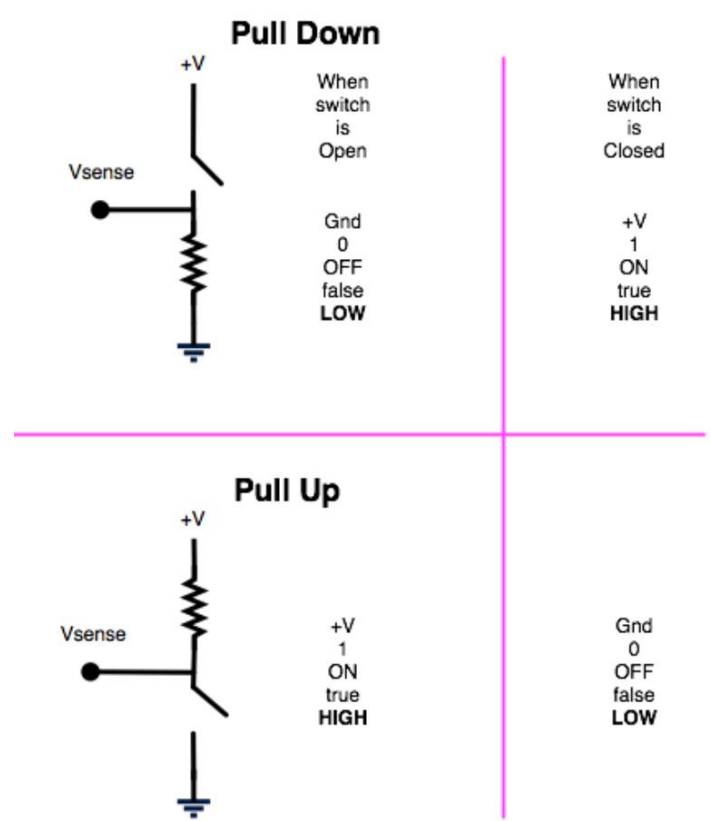


Arduino: Brain, make decisions

Motor driver: Voltage switching, Current amplification, Direction control

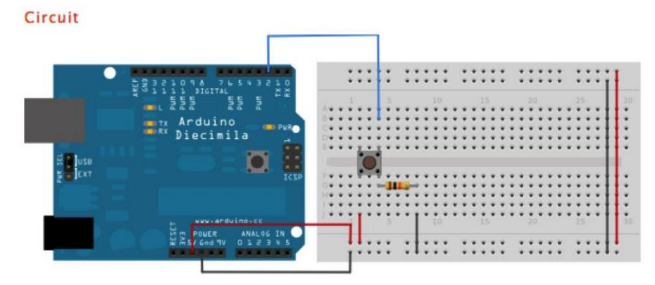


Motor



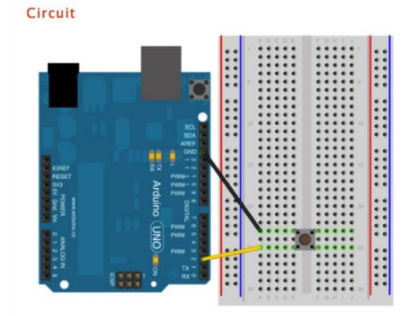
Pull-down

<http://arduino.cc/en/Tutorial/Button>

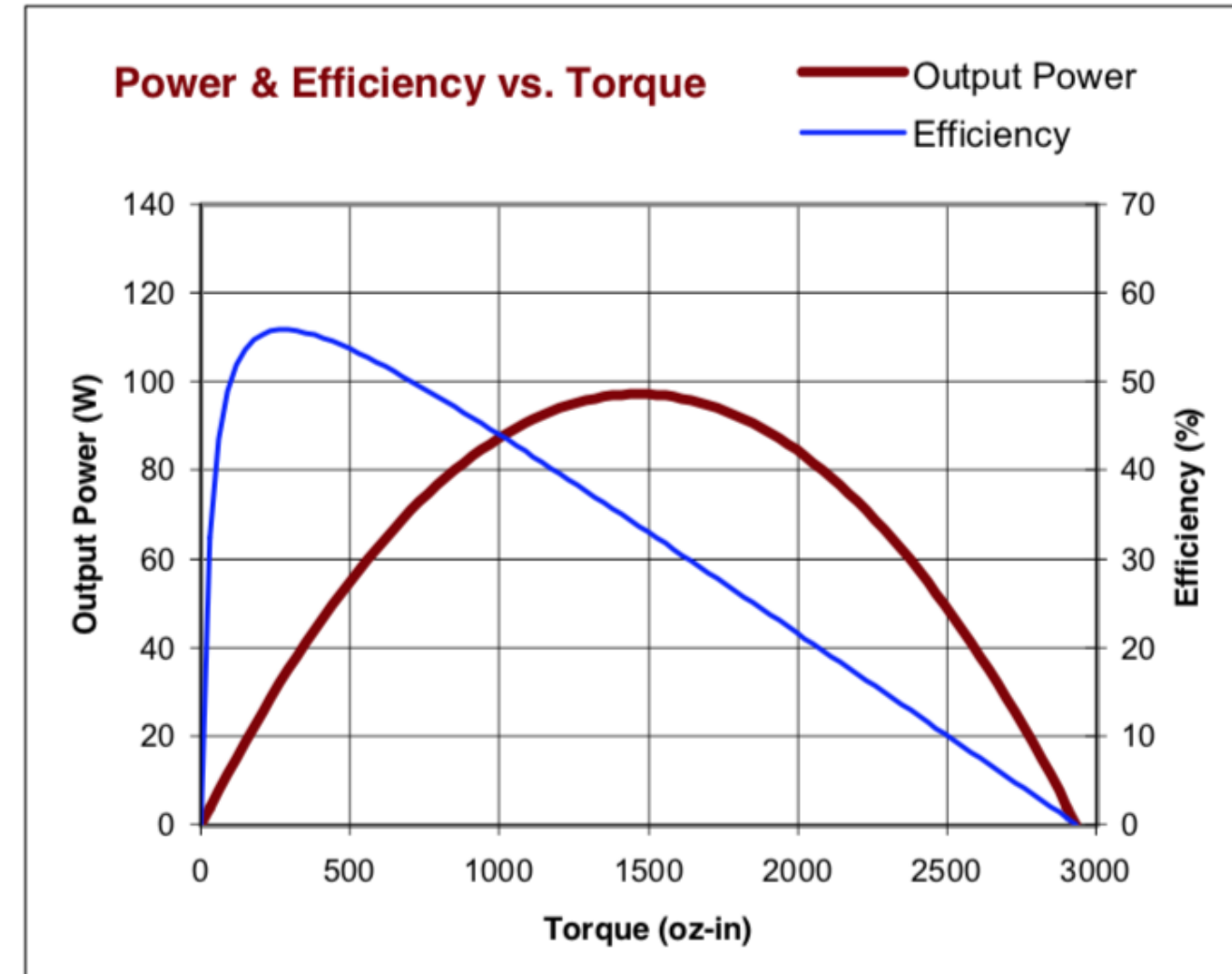
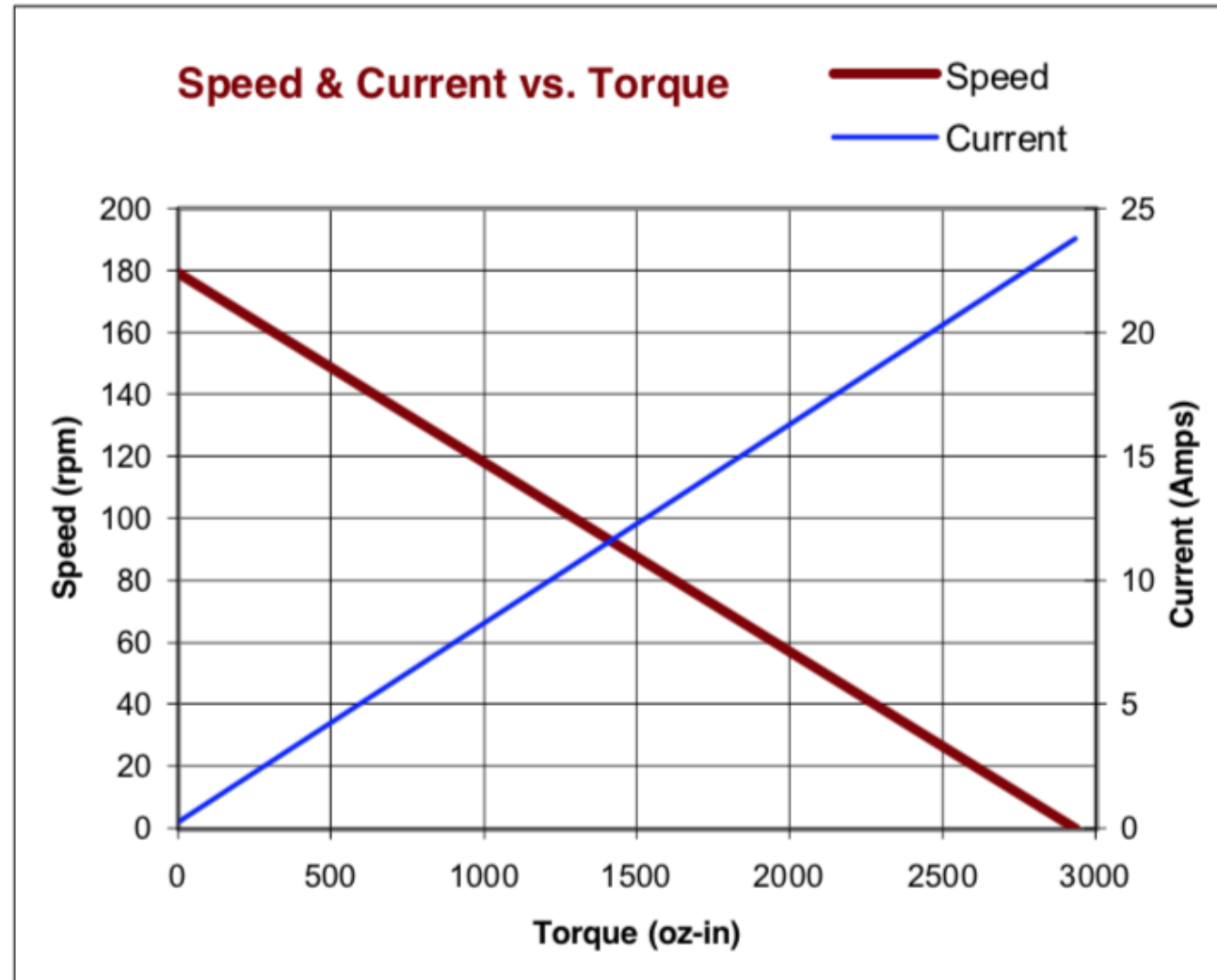


Pull-up

<http://arduino.cc/en/Tutorial/InputPullupSerial>



Last Time: Motor speed & power vs torque



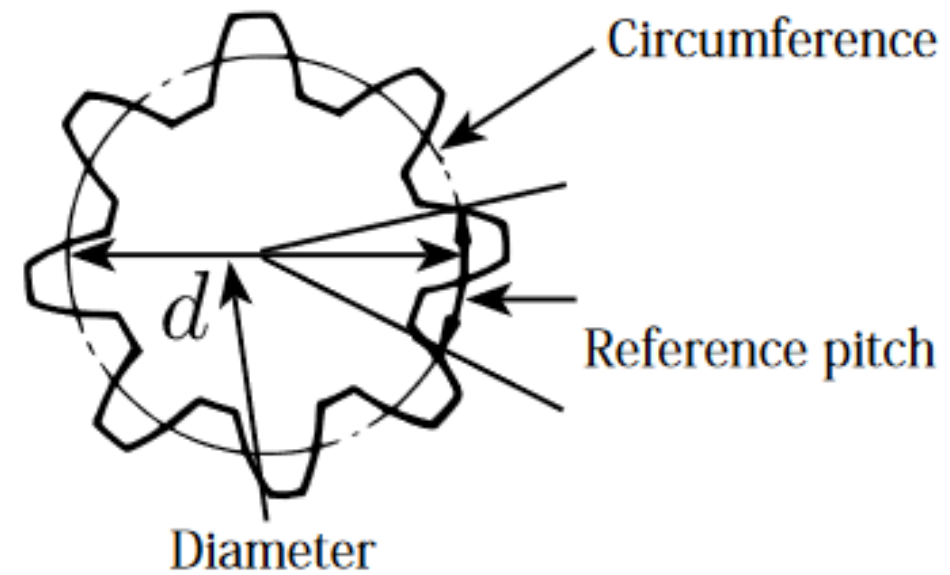
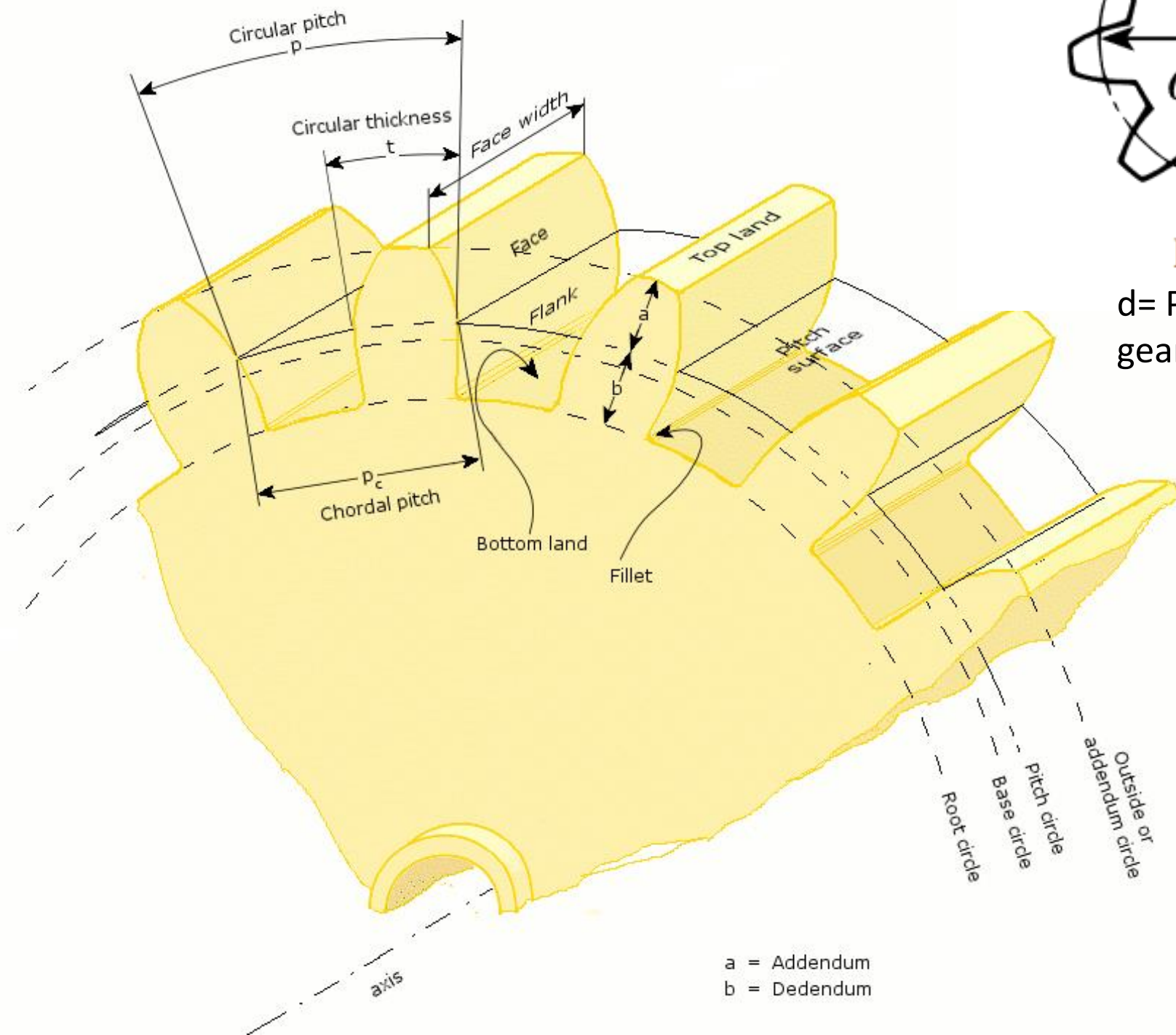
All values are nominal. Specifications subject to change without notice. Graphs are shown for reference only.

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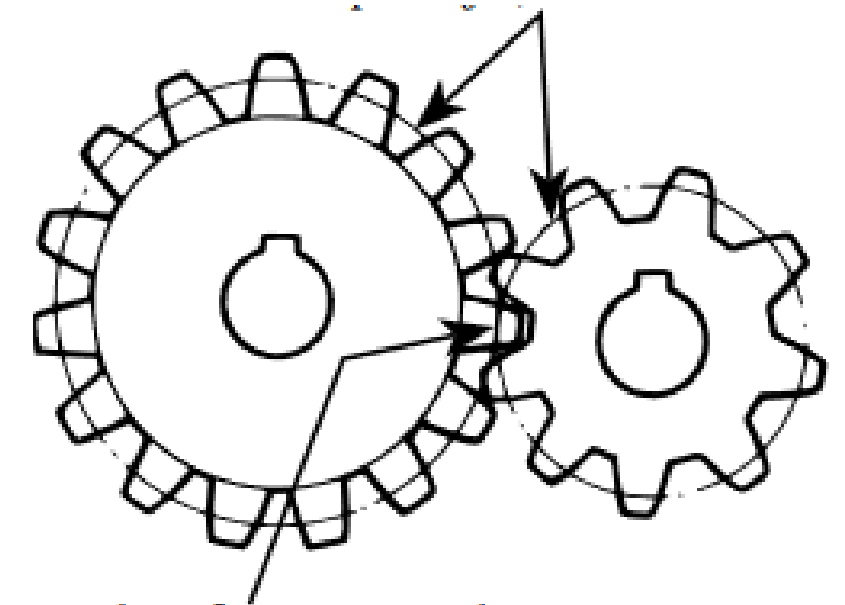
Why do we use gears?

- Changing Speed and Torque
- Changing Direction of Motion
- Mechanical Advantage and Efficiency
- Synchronization and Control
 - Gears ensure parts move in a synchronized, predictable way.
- Compact and Durable Power Transmission
 - Gears can transmit large amounts of power in a relatively small space.

Gear anatomy



d = Pitch diameter, circle where gears make contact.



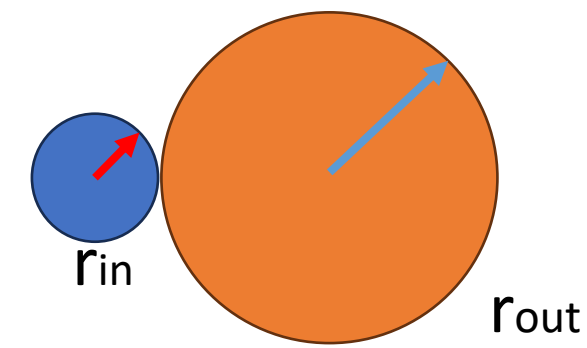
Both reference circles come in contact

$$\text{Pitch} = \frac{\text{Circumference}}{\text{Number of teeth}} = \frac{\pi d}{N}$$

$$m = \frac{\text{Pitch}}{\pi}$$

Only gears with the same pitch or module(m) can mesh correctly.

Gear ratio



When two gears mesh, the arc length at the contact point is the same:

$$r_{\text{in}} \theta_{\text{in}} = r_{\text{out}} \theta_{\text{out}}$$

- r : radius
- θ : rotation
- ω : angular velocity
- α : angular acceleration

Therefore:

$$r_{\text{in}} \omega_{\text{in}} = r_{\text{out}} \omega_{\text{out}}$$

Power Conservation (Approximate)

$$P \approx \text{constant} \Rightarrow \tau_{\text{in}} \omega_{\text{in}} = \tau_{\text{out}} \omega_{\text{out}}$$

The gear ratio (GR) can be expressed in multiple equivalent forms:

$$GR = \frac{r_{\text{out}}}{r_{\text{in}}} = \frac{N_{\text{out}}}{N_{\text{in}}} = \frac{\omega_{\text{in}}}{\omega_{\text{out}}} = \frac{\tau_{\text{out}}}{\tau_{\text{in}}}$$

Where:

- N : number of teeth
- ω : angular velocity
- τ : torque

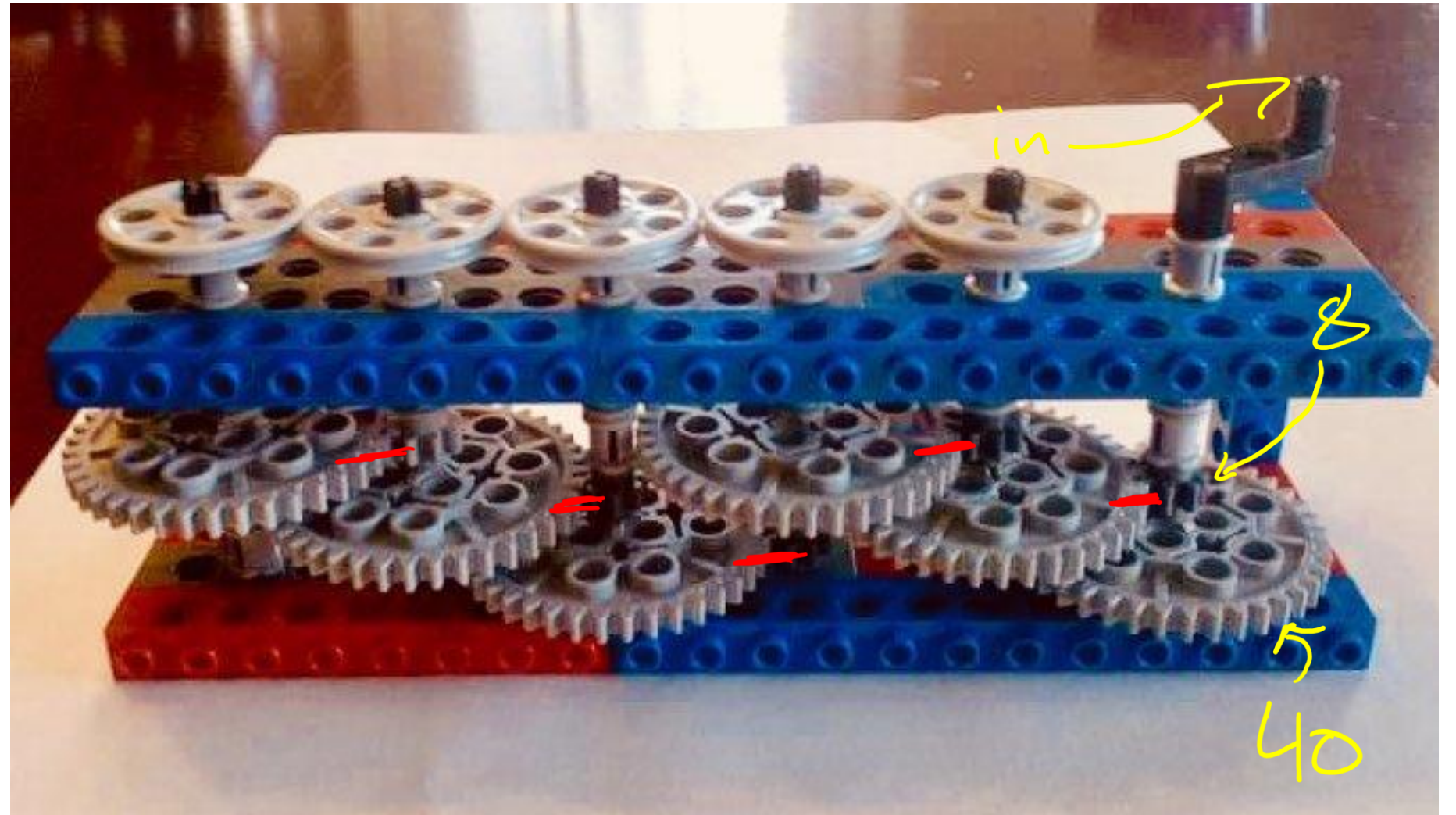
Physical Meaning

A **larger gear** rotates slower but delivers **more torque**.

A **smaller gear** rotates faster but delivers **less torque**.

Gear trains

- A series of gears meshed together
- Overall gear ratio is the product of the individual gear ratios

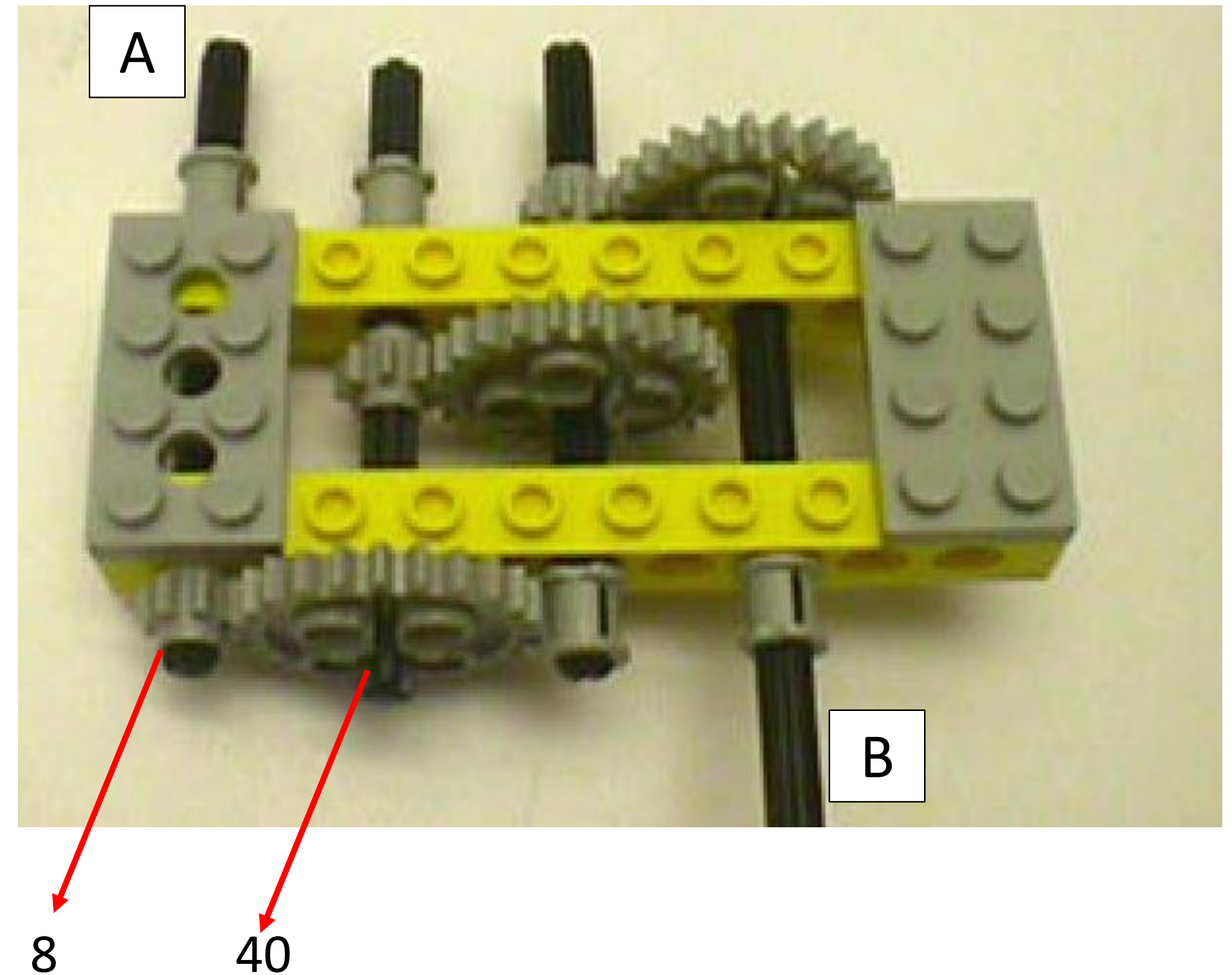


$$GR = \frac{40}{8} * \frac{40}{8} * \frac{40}{8} * \frac{40}{8} * \frac{40}{8} = \frac{5^5}{1} = \frac{3125}{1}$$



If the motor drives shaft A with a torque of **500 oz-in** and a speed of **100 rpm**, what is the torque and speed of shaft B?

$$GR = 40/8$$

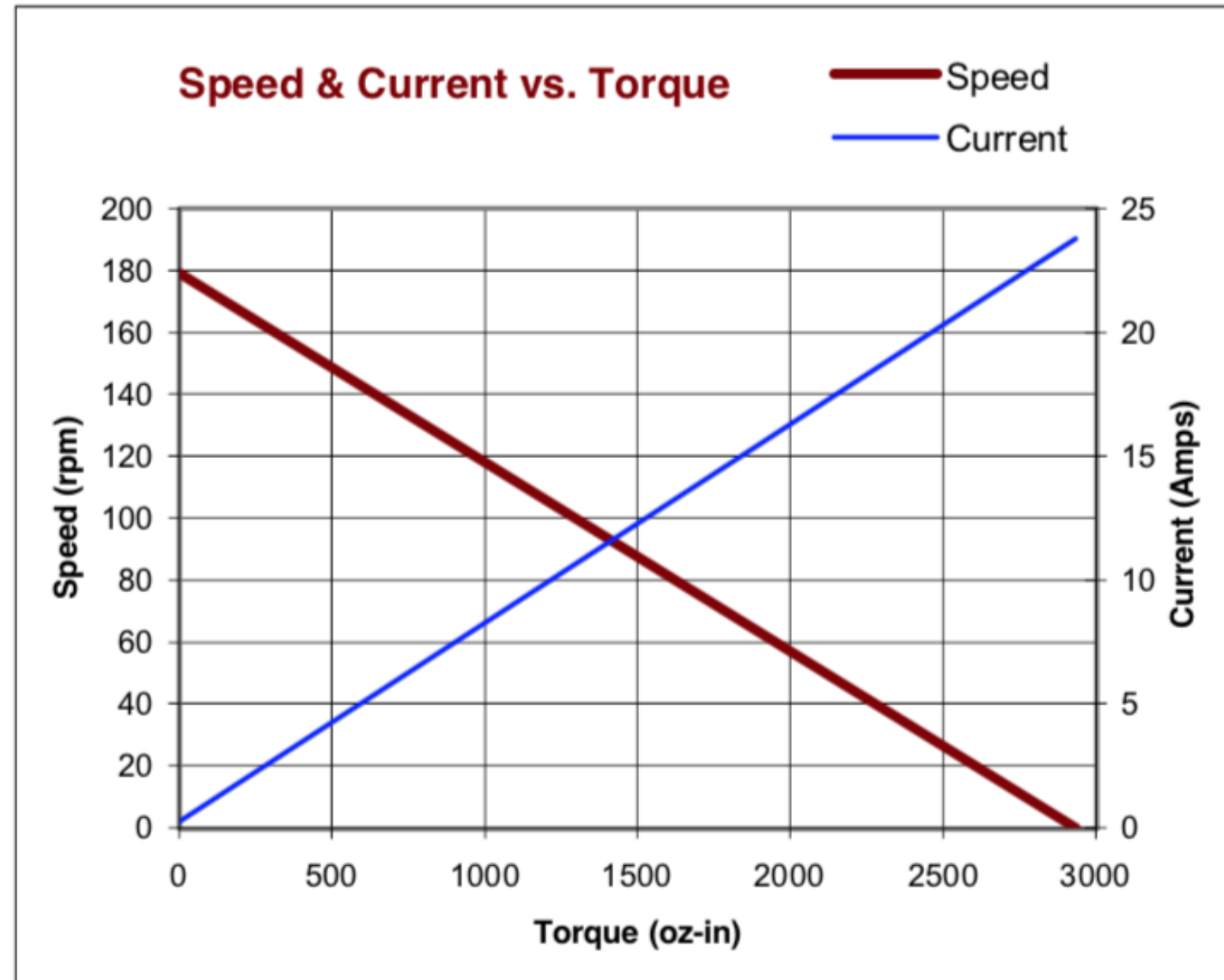


$$GR = \frac{40}{8} \cdot \frac{40}{8} \cdot \frac{40}{8} = \frac{125}{1} = \frac{5^3}{1}$$

$$\tau_{out} = GR \cdot \tau_{in} = 62500 \text{ oz-in}$$

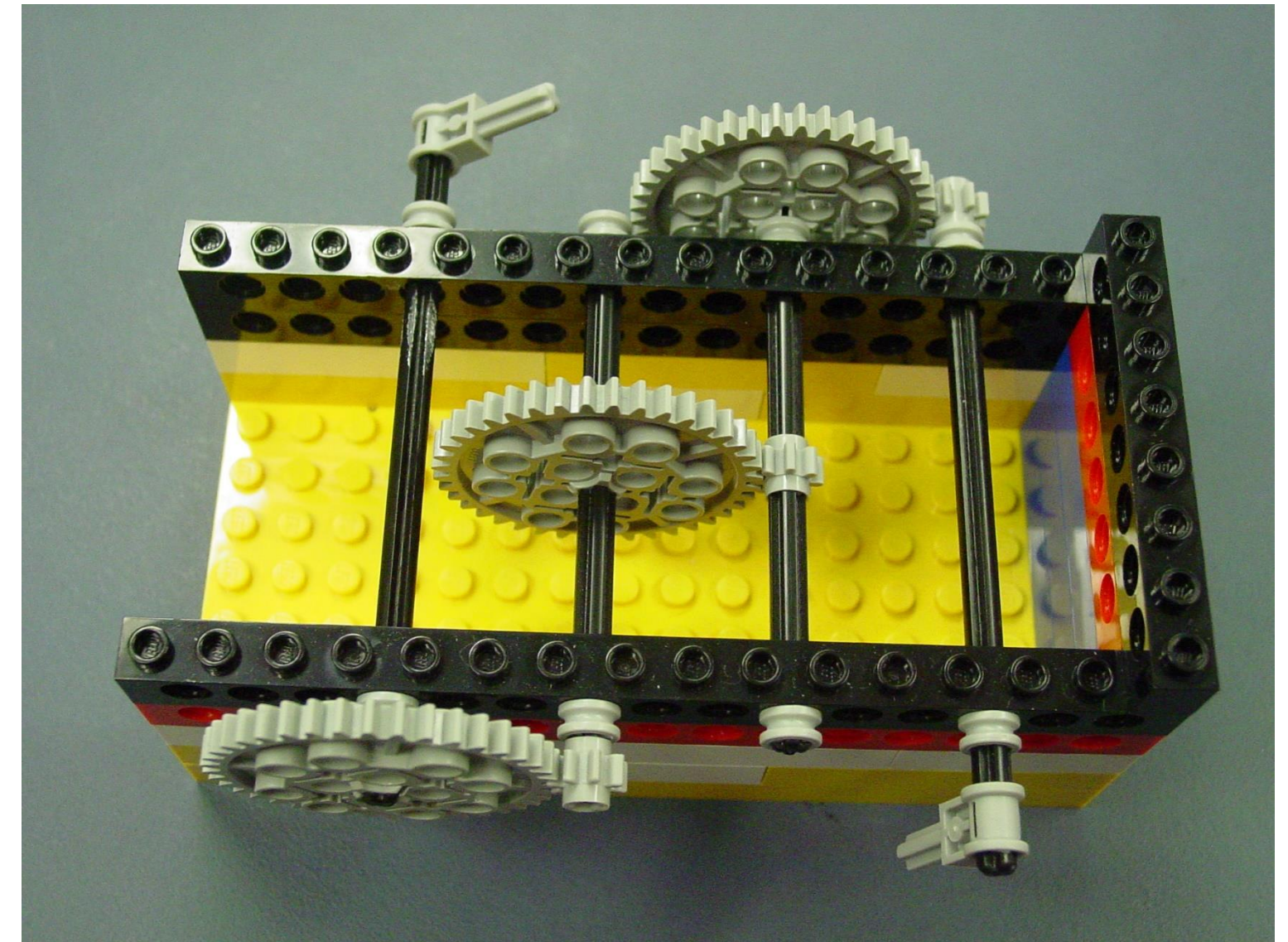
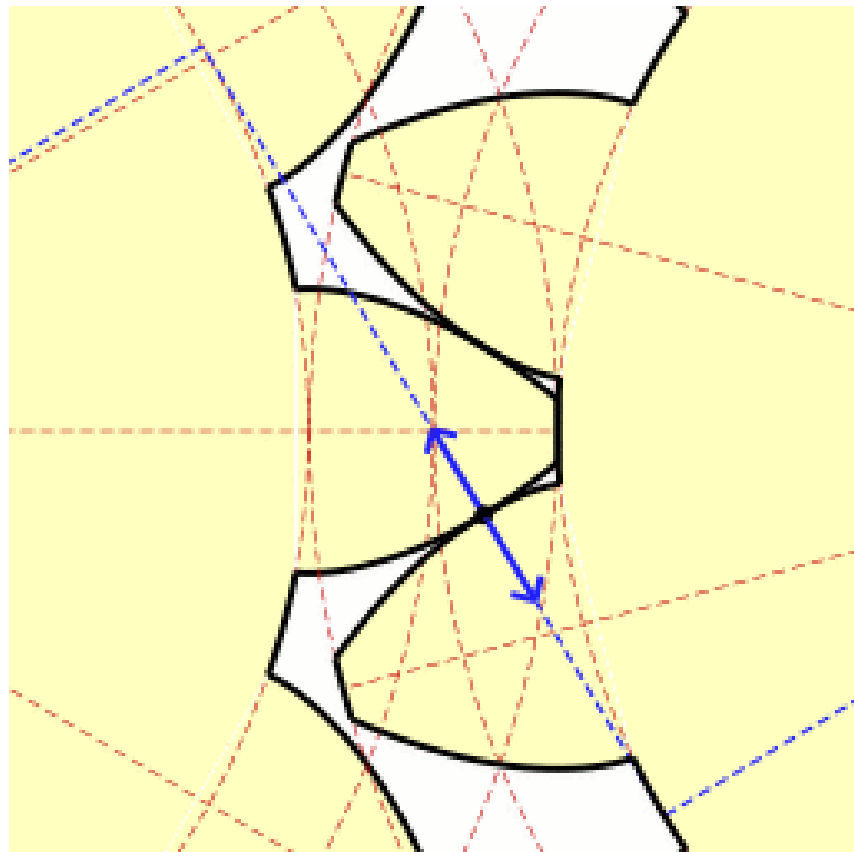
$$\omega_{out} = \frac{\omega_{in}}{GR} = 0.8 \text{ rpm}$$

Relationship to motor



Gear types: spur gears

- Most common
- Mesh with parallel axes
- *Involute* tooth profile



<http://cs.smith.edu/~100e/gears.html>

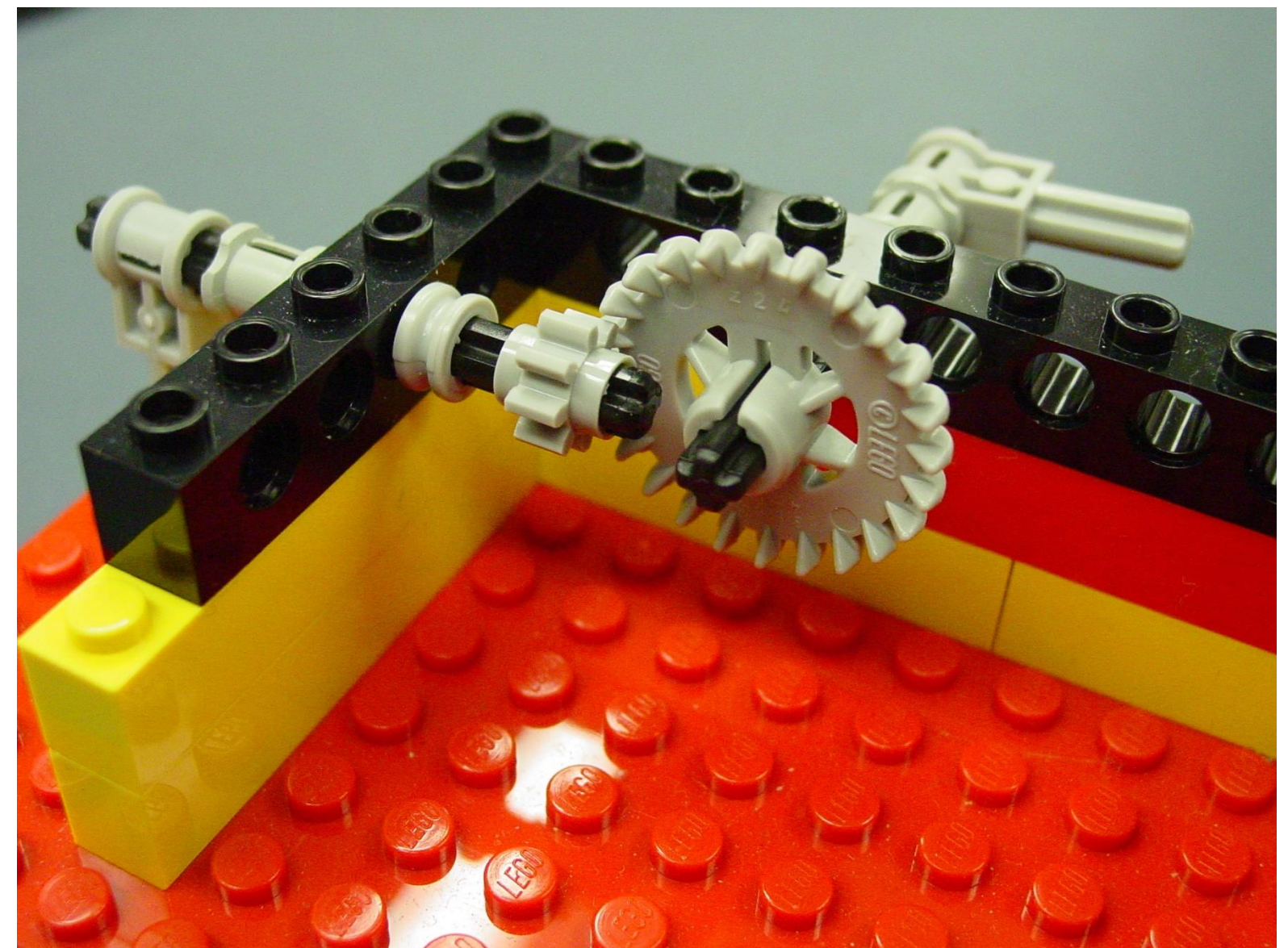
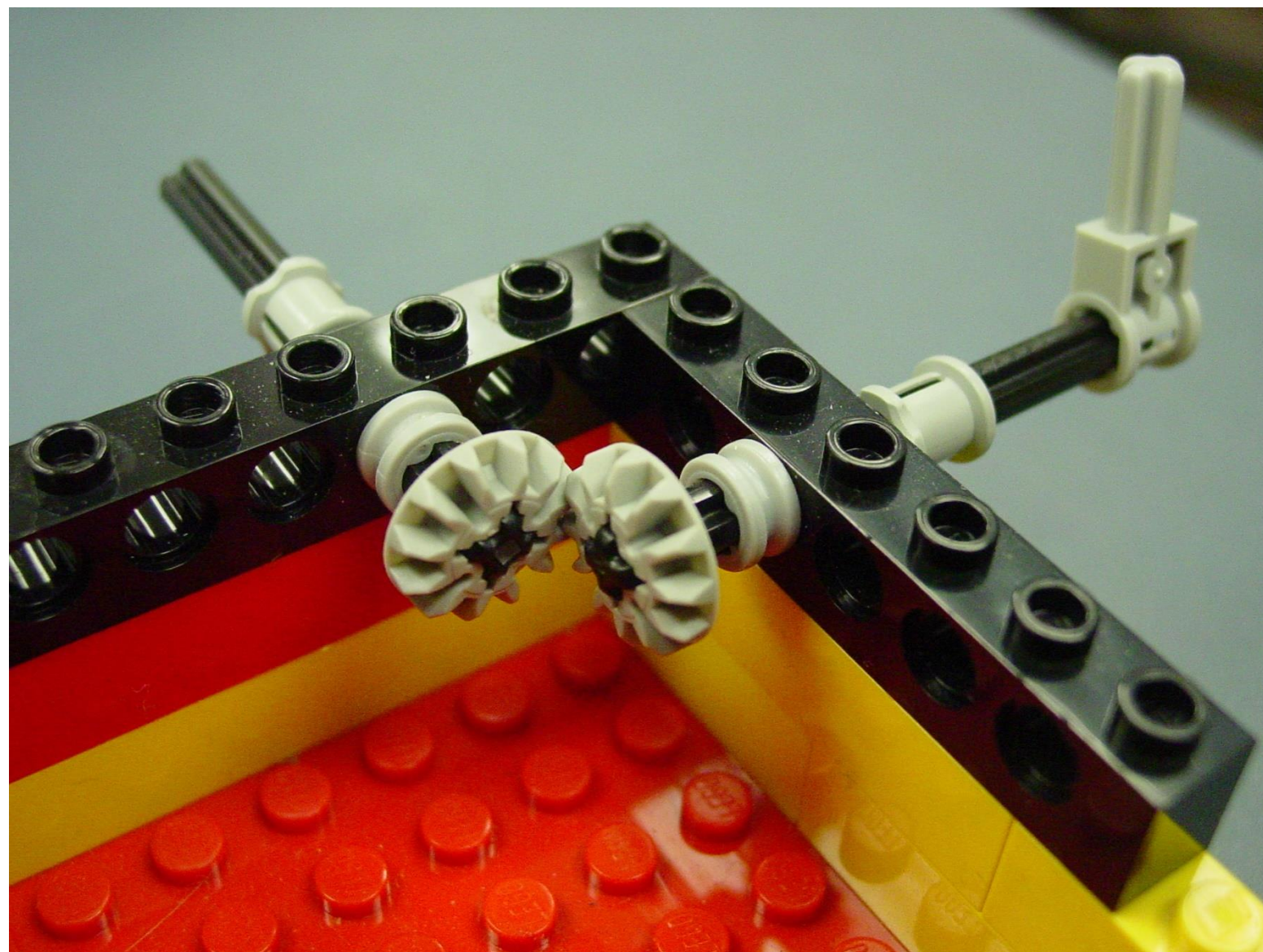
Gear Types: Helical Gears

- Curved gear produces gradual contact and release
- Better for high power transmission and lower noise
- Can have parallel or perpendicular axes



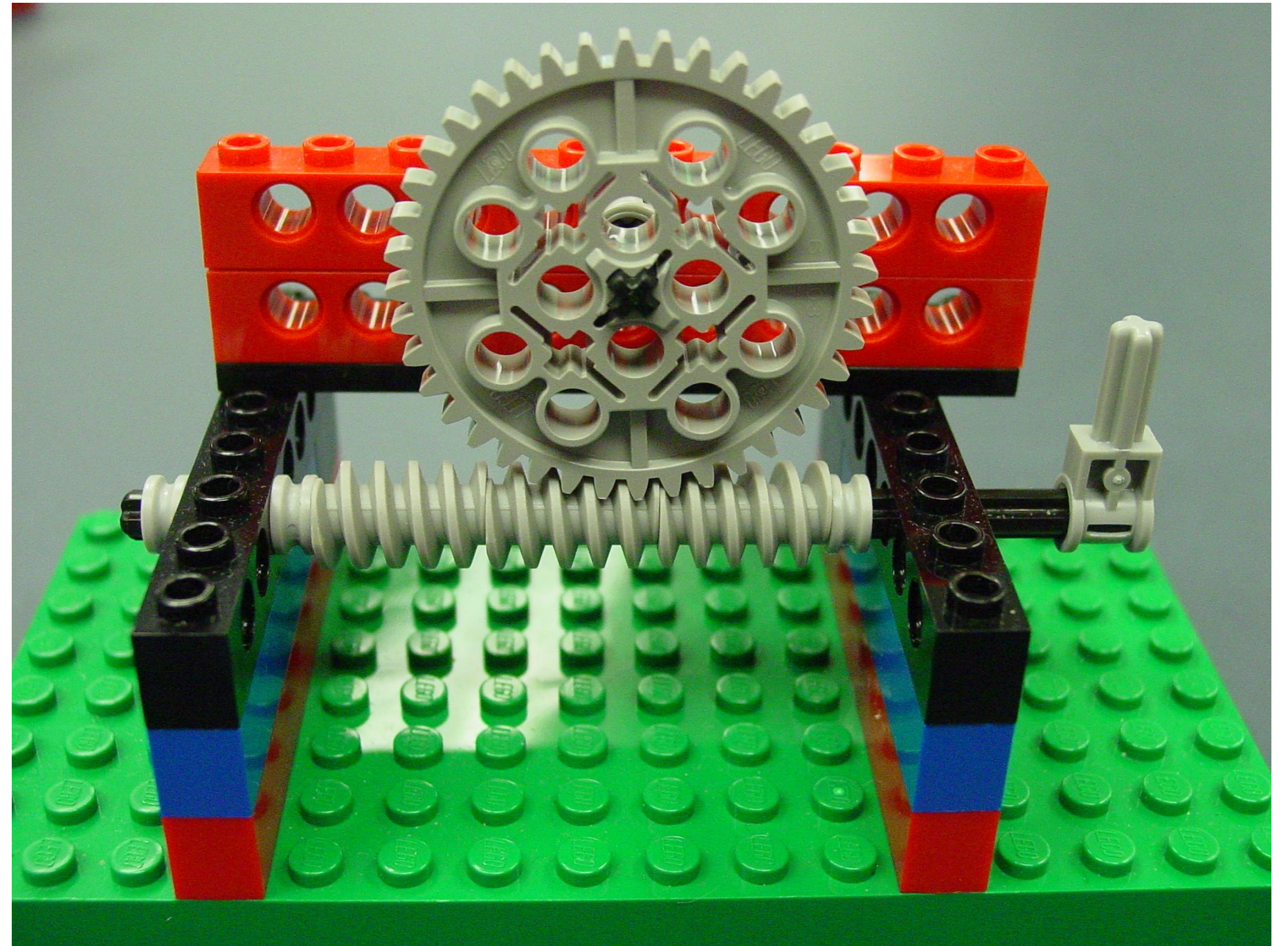
Gear Types: Bevel and Crown

- Used to change directions



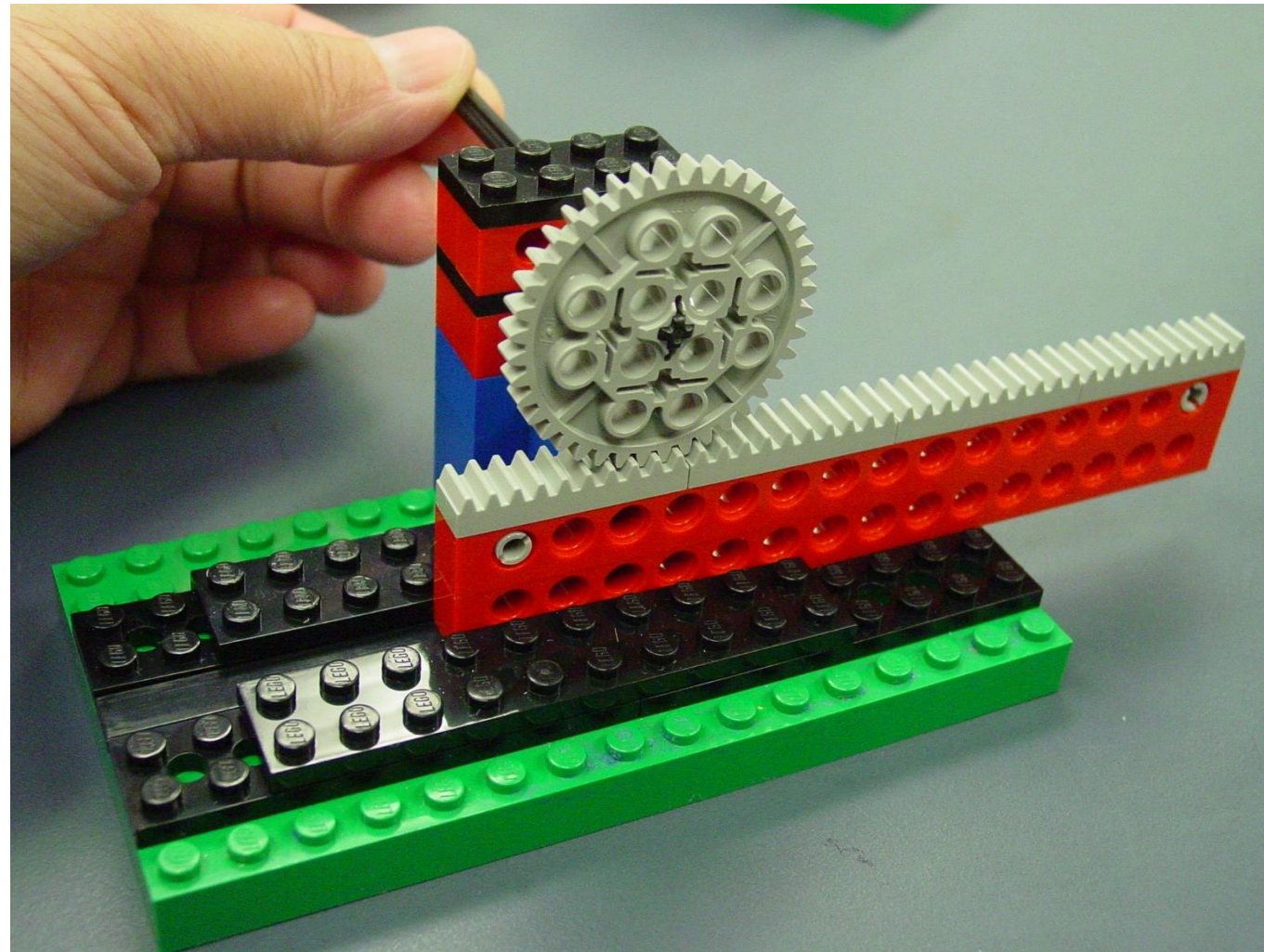
Gear Types: Worm Gear

- High torque capabilities
- One rotation moves 1 tooth length
 - $\frac{\text{Number of teeth}}{1}$ gear ratio



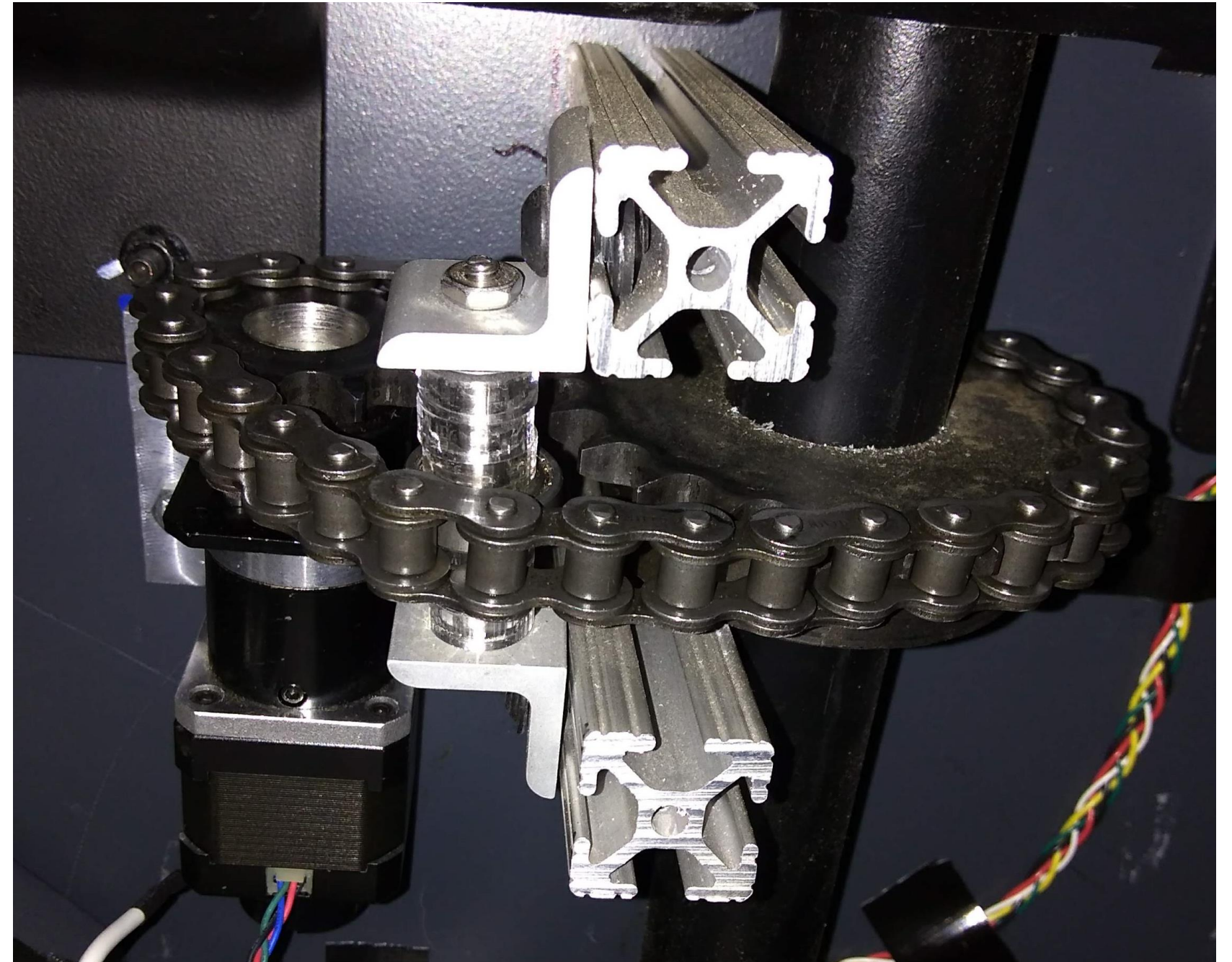
Gear Types: Rack and Pinion

- Converts rotation into translation



Chains/Pulleys

- Chains
 - Allows distant actuation
 - Gears don't have to be in direct contact
 - Better for high torque, not good for high speed
- Pulleys
 - No teeth, use belts
 - Don't need to match teeth
 - Arbitrary diameter
 - Relies on friction

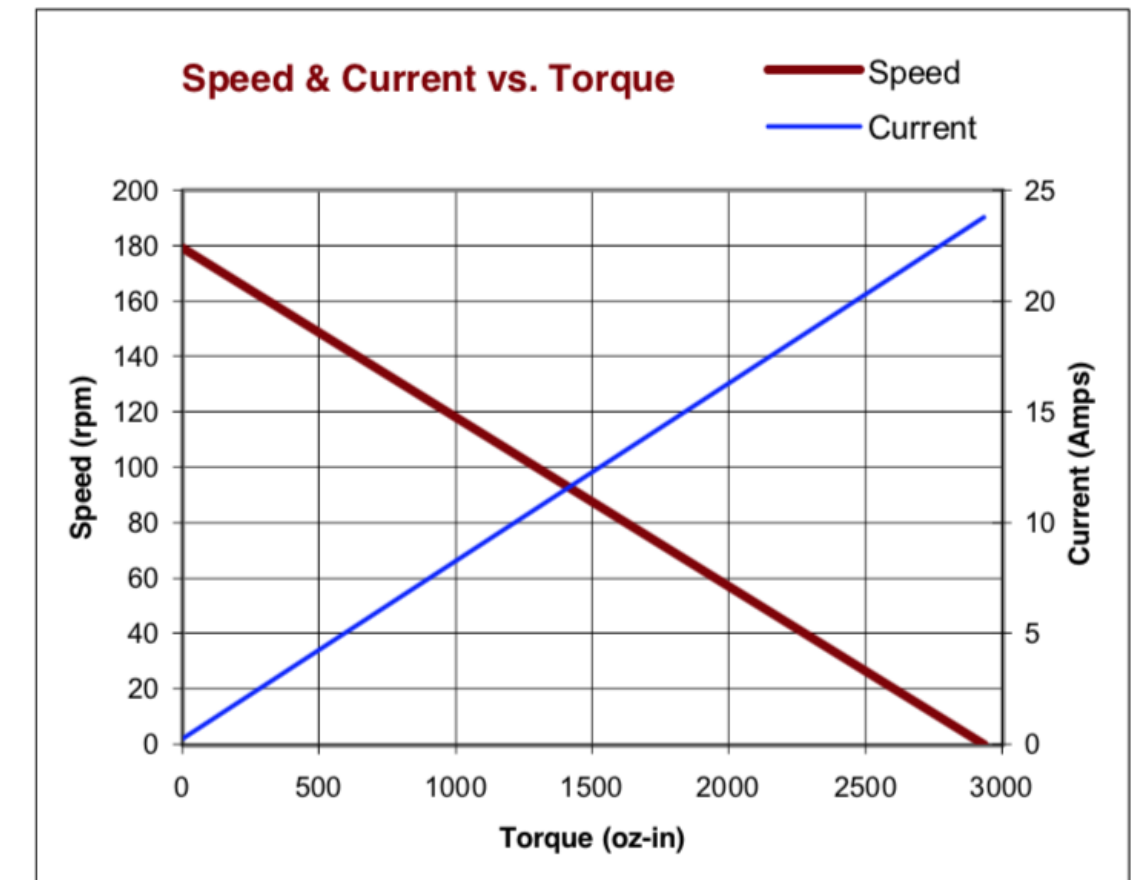


Gear Train Design Tips

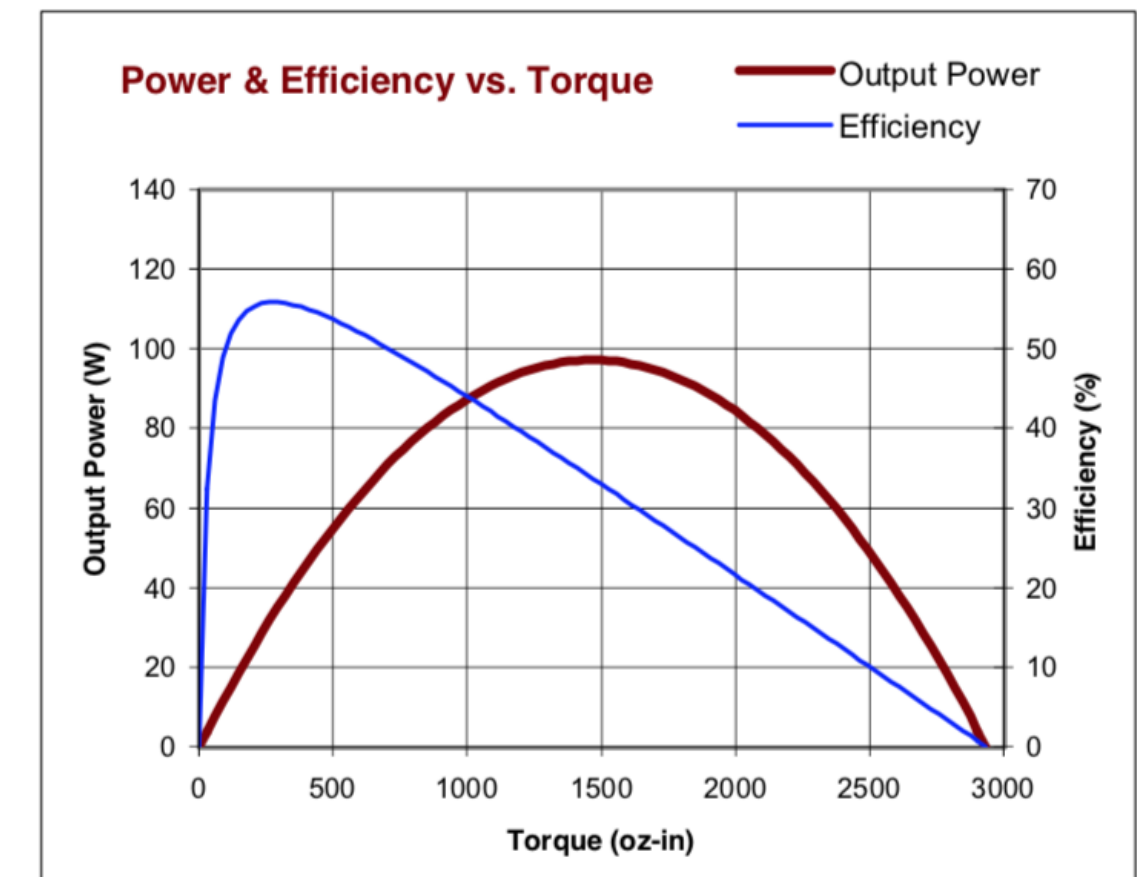
- Work backward from the final drive
 - Figure out where the wheels go first. Motor location is flexible
- Consider tire size when thinking about gear trains, $v = r\omega$
- If the gear train is performing badly
 - Make sure the stop bushings aren't squeezing too tightly
 - Check that all pieces are square and aligned
 - To test a gear train, **try driving it backward**

Lab 1 – Motor characterization

- Use your motors to lift a weight
- Measure average motor speed by timing the process of lifting
 - Generate torque-speed curve and torque-power curve
- Measure current and voltage to calculate electrical power
- Measure speed variation by varying pwm
- Measure reliability by repeating a trial



All values are nominal. Specifications subject to change without notice. G1



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